YeonJun Choi

PhD Student · Applied Physics

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Education

Ph.D. in Applied Physics — Aug. 2022 – present
Texas A&M University · College Station, TX, USA

- Advisor: Prof. Joseph H. Ross, Jr.
- NMR/NQR studies of condensed matters and quantum materials
- Transferred from the Department of Chemical Engineering

M.S. in Physics — Mar. 2019 – Aug. 2022
Hankuk University of Foreign Studies · Yongin, South Korea

- Advisor: Prof. Bowha Lee
- Dissertation: *Effect of particle size and CIP addition on magnetic properties and core loss in high frequency bands of Fe–Si–Cr alloys*
- Military service: Dec. 2019 – Aug. 2021

Intensive English Language Course — Jan. 2017 – Feb. 2017
University of Delaware · Newark, DE, USA

B.S. in Electronic Physics — Mar. 2015 – Feb. 2019
Hankuk University of Foreign Studies · Yongin, South Korea

- Undergraduate research advisor: Prof. Bowha Lee

Research Experience

Graduate Research Assistant — Jan. 2024 – present
Texas A&M University · College Station, TX

- Advisor: Prof. Joseph H. Ross, Jr.
- Computational calculations and exponential fittings for quadrupole and magnetic T_1 and T_2 in solid-state NMR
- NMR/NQR studies of Indium intercalated 2H-TaS_2 transition metal dichalcogenides (TMDCs)
- NMR/NQR studies of an altermagnetic material candidate, Vanadium intercalated 2H-TaS_2 TMDCs
- Supported by Texas A&M University

Graduate Research Assistant — Jan. 2025 – Aug. 2025
Texas A&M University · College Station, TX

- Advisor: Prof. Jonas Karthein
- Commissioned Collision-Induced Dissociation Gas Cell (CID-GC) and conducted precision laser spectroscopy of atoms and molecules containing rare isotopes
- Developed piezoelectric-based laser ablation ion source for radioactive molecular ions
- Supported by Texas A&M University

Graduate Research Assistant — Jun. 2020 – Aug. 2022
Hankuk University of Foreign Studies · Yongin, South Korea

- Advisor: Prof. Bowha Lee
- Development of metallic soft materials and magnetic components with excellent high-frequency (500 kHz) loss characteristics for 5G and next-generation communication
- Research on EMI shielding materials for multi-bands
- Supported by the National Research Foundation of Korea and Gyeonggi Province

Undergraduate Research Assistant Mar. 2018 – Feb. 2020
Hankuk University of Foreign Studies · Yongin, South Korea

- Advisor: Prof. Bowha Lee
- Investigated millimeter-wave absorption and shielding of polyaniline/Z-type hexaferrite nanocomposites
- Supported by the National Research Foundation of Korea

Publications

Total: 11 | First-author: 5 | Citations: 159

In Preparation

1. **Choi, Y.J.**, Li, R., & Ross, J.H. "Quadrupole and magnetic T_1 and T_2 in solids NMR: Combined behavior and exponential fitting." *In preparation*, 2026.

Published

1. Kim, S.W., **Choi, Y.J.**, Kim, Y.R., Kim, D.H., In, Y.J., Kim, T.H. & Lee, B.W. "Permeability modulation of Fe-Ni/nanoparticle (Ni, Zn) soft magnetic composites." *Current Applied Physics, 69, 42–46*, 2025.

2. **Choi, Y.J.**, Kim, S.W., Phan, T.L., Lee, B.W., & Yang, D.S. "Tetragonal-structural changes influenced the magnetic and ferroelectric properties of (Y, Fe)-codoped $BaTiO_3$ ceramics." *Current Applied Physics, 53, 39–45*, 2023.

3. **Choi, Y.J.**, Kim, D.H., & Lee, B.W. "AC magnetic loss reduction of Fe-(x)Si soft magnetic composites." *Journal of Superconductivity and Novel Magnetism, 36, 909–914*, 2023.

4. **Choi, Y.J.**, Lee, M.Y., & Lee, B.W. "Magnetic property improvement and core-loss reduction of Fe–Si–Cr-based soft magnetic composites with addition of Fe-50Ni nanopowder." *The Journal of The Minerals, Metals & Materials Society, 75, 1261–1269*, 2023.

5. Lee, M.Y., **Choi, Y.J.**, Woo, H.J., Kim, S.W., & Lee, B.W. "Microwave absorption properties of $Ni_{0.6}Zn_{0.4}Fe_2O_4$ composites with nonmagnetic nanoferrite percentages." *Ceramics International, 48(14), 20187–20193*, 2022.

6. Lee, M.Y., **Choi, Y.J.**, Lee, S.H., Ahn, J.H., & Lee, B.W. "Controlling properties of metal–polymer soft magnetic composites through microstructural deformation for power inductor applications." *Journal of Materials Science: Materials in Electronics, 33, 15763–15772*, 2022.

7. **Choi, Y.J.**, Ahn, J.H., Kim, D.H., Kim, Y.R., & Lee, B.W. "Core-loss reduction of Fe–Si–Cr crystalline alloy according to particle size in the high frequency band." *Current Applied Physics, 39, 324–330*, 2022.

8. Kim, D.H., Yeo, J.G., **Choi, Y.J.**, Lee, S.H., An, S.Y., Kim, J.Y., & Lee, B.W. "Magnetic properties of amorphous metallic composites with various particle sizes." *Journal of Korean Physical Society, 79, 1037–1041*, 2021.

9. **Choi, Y.J.**, Ahn, J.H., Kim, S.W., Kim, Y.R., & Lee, B.W. "Improvement in power inductor performance at 3 MHz by mixing carbonyl iron powder with Fe–Si–Cr crystalline alloy." *MRS Communications, 11, 457–461*, 2021.

10. Tran, N., **Choi, Y.J.**, Phan, T.L., Yang, D.S., & Lee, B.W. "Electronic structure and magnetic and electromagnetic wave absorption properties of $BaFe_{12-x}Co_xO_{19}$ M-type hexaferrites." *Current Applied Physics, 19(12), 1343–1348*, 2019.

11. Yeo, J.G., Kim, D.H., **Choi, Y.J.**, & Lee, B.W. "Improving power-inductor performance by mixing sub-micro Fe powder with amorphous soft magnetic composites." *Journal of Electronic Materials, 48, 6018–6023*, 2019.

Presentations

1. **Choi, Y.J.**, Li, R., & Ross, J.H. Practical T_1 Analysis for Quadrupolar Solid-State NMR Experiments. *US-Korea Conference 2026 (UKC 2026)*, Orlando, FL, Aug. 2026. [Poster]

2. **Choi, Y.J.**, Gamage, N., Ringle, R., Schnoor, N., Zhou, Y. & Karthein, J. The Making and Breaking of Radioactive Molecules. *Low-Energy Community Meeting 2025 (LECM 2025)*, College Station, TX, Aug. 2025. [Poster]

3. **Choi, Y.J.**, Ringle, R., Gamage, N., Schnoor, N. & Karthein, J. The Development of Collision-Induced Dissociation Gas Cell (CID-GC) for Quantum Sensing of Nuclei. *US-Korea Conference 2025 (UKC 2025)*, Atlanta, GA, Aug. 2025. [Poster]
4. **Choi, Y.J.**, Ringle, R., Gamage, N., Zhou, Y., Schnoor, N. & Karthein, J. The Making and Breaking of Radioactive Molecules. *Exotic Beam Summer School 2025 (EBSS 2025)*, Berkeley, CA, Jun. 2025. [Poster]
5. **Choi, Y.J.**, Yu, S.C. & Lee, B.W. Effect of adding various powders with small particles on the magnetic properties and core-loss reduction of soft magnetic composites. *19th International Conference on Advanced Nanomaterials*, Aveiro, Portugal, Jul. 2022. [Poster & Oral]
6. **Choi, Y.J.**, Lee, M.Y., Kim, S.W. & Lee, B.W. Microwave absorption properties of FeSiCr soft magnetic composites depending on particle sizes. *2022 Korean Magnetic Society Summer Meeting*, Jeju, Korea, May. 2022. [Poster]
7. **Choi, Y.J.**, Lee, M.Y., Jeong, W.H., Phan, T.L. & Lee, B.W. Influence of Y/Fe Co-doping on Crystal Structure and Magnetic Properties of $BaTiO_3$. *2022 Korea Dielectric Symposium*, Busan, Korea, Feb. 2022. [Poster]
8. **Choi, Y.J.**, Ahn, J.H., Kim, S.W., Kim, D.Y. & Lee, B.W. The effect of particle size on core-losses of Fe–Si–Cr soft magnetic composites. *2022 Joint MMM-Intermag Conference*, New Orleans, LA, Jan. 2022. [Poster]
9. **Choi, Y.J.**, Ahn, J.H., Lee, M.Y., Kim, D.Y. & Lee, B.W. Improvement in Q-factor and Resonance Frequency Depending on the Particle Sizes of Fe–Si–Cr Alloy. *The 12th International Conference on Advanced Materials and Devices*, Jeju, Korea, Dec. 2021. [Poster]
10. **Choi, Y.J.**, Lee, M.Y., Woo, H.J., Cho, K.W. & Lee, B.W. Permeability Dependence on Particle Size of Alloy Based FeSiCr Metal Powder. *2019 Korean Magnetic Society Winter Meeting*, Jeju, Korea, Nov. 2019. [Poster]
11. Lee, M.Y., **Choi, Y.J.**, Tran, N., Ahn, J.H. & Lee, B.W. Fabrication of Permanent Magnet by 3D Printing Technique. *64th Annual Conference on Magnetism and Magnetic Materials*, Las Vegas, NV, Nov. 2019. [Poster]
12. **Choi, Y.J.** A Study on the Magnetic Characteristics of Powder Inductor with Metal Nanopowder. *2019 Korea-Vietnam Symposium on Magnetic Materials*, Danang, Vietnam, Jul. 2019. [Oral]
13. Tran, N., **Choi, Y.J.**, Lee, M.Y., Ahn, J.H., Phan, T.L. & Lee, B.W. Structural, magnetic and electromagnetic wave absorption properties of La-BaM/PANI nanocomposites. *Nanotech France 2019*, Paris, France, Jun. 2019. [Poster]
14. **Choi, Y.J.**, Tran, N., Phan, T.L. & Lee, B.W. Structural and magnetic properties of $Mn_{1+b}Fe_{2-b}O_{4-\delta}$ NPs: A view from a combination of EDS and XAS. *2019 Korean Magnetic Society Summer Meeting*, Busan, Korea, May. 2019. [Poster]

Teaching Experience

Term	Course	Institution
Fall 2026	Newtonian Mechanics for Engineering & Science (PHYS 206) — TA	TAMU
Summer 2026	Electricity & Magnetism for Students in Engineering & Science (PHYS 207) — TA	TAMU
Spring 2026	Newtonian Mechanics for Engineering & Science (PHYS 206) — TA	TAMU
Summer 2024	College Physics (PHYS 201) — TA	TAMU
Spring 2024	Newtonian Mechanics for Engineering & Science (PHYS 206) — TA	TAMU
Spring 2023	Chemical Engineering Heat Transfer Operations (CHEN 323) — TA	TAMU
Fall 2022	Chemical Engineering Materials (CHEN 322) — TA	TAMU
Spring 2022	General Physics Laboratory 1 — TA	HUFS
Fall 2019	Modern Physics Laboratory 2 — TA	HUFS
Spring 2019	Modern Physics Laboratory 1 — TA	HUFS
Fall 2018	Modern Physics Laboratory 2 — TA	HUFS
Spring 2018	General Physics Laboratory 1 — TA	HUFS

Awards, Fellowships & Grants

Aug. 2026	UKC 2026 Travel Support, Korean-American Scientists and Engineers Association	$200
Aug. 2025	UKC 2025 Travel Support, Korean-American Scientists and Engineers Association	$200
Oct. 2024	Encouragement Award, The Korean Physical Society	KRW 700,000
Mar. 2022	HUFS Departmental Scholarship, Hankuk University of Foreign Studies	KRW 4,801,000
Dec. 2021	IEEE Magnetics Society Travel Grant, 2022 Joint MMM-Intermag Conference	$1,000
Sep. 2021	Global Talent Scholarship, Hankuk University of Foreign Studies	KRW 4,999,000

Skills

Materials Fabrication Chemical Vapor Transport (CVT), Solid-state reaction, co-precipitation, pulsed wire evaporation (PWE), ball-milling

Characterization Solid-state NMR spectrometer, X-ray Diffractometer (XRD), Vibrating Sample Magnetometer (VSM), impedance analyzer, B–H analyzer, network analyzer, ferroelectric tester, Raman spectroscopy, FT-IR

Computational Wien2k DFT calculation, Mathematica, Python, LaTeX, FullProf, OriginPro, Veusz, AutoCAD, Microsoft Office

Professional Memberships

American Physical Society (APS) · The Korean Physical Society (KPS) · The Korean Magnetics Society (KMS) · Korean-American Scientists and Engineers Association (KSEA)

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